IndiaWorks Unified SaaS - Technical Architecture & Data Design

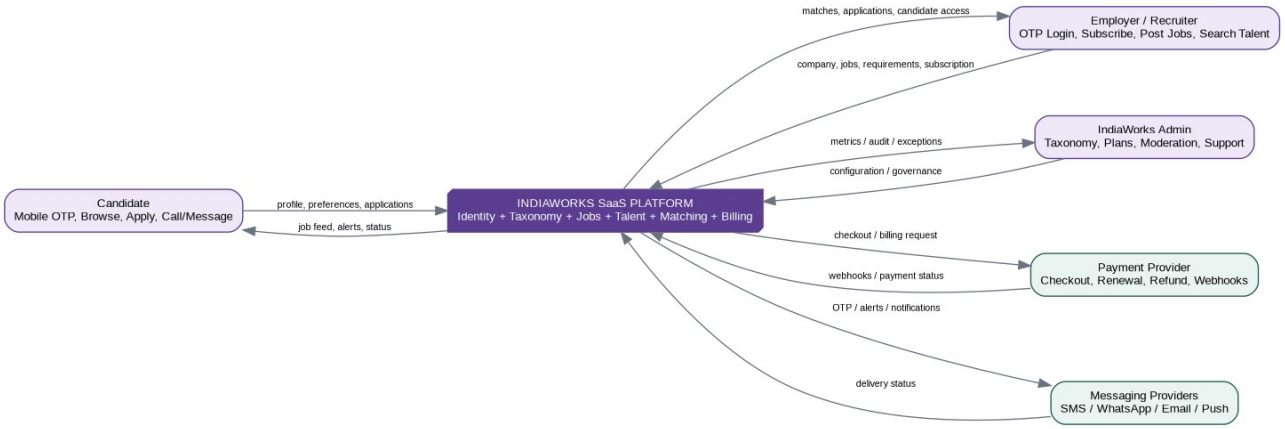
DFD, database domains, GraphQL, entitlements, search, security and deployment

IndiaWorks | Unified SaaS Architecture v7
Generated 25 September 2026
Baseline: v5 Unified SaaS + v6 Product/Technical Review + v7 Five-Candidate Commercial Gate

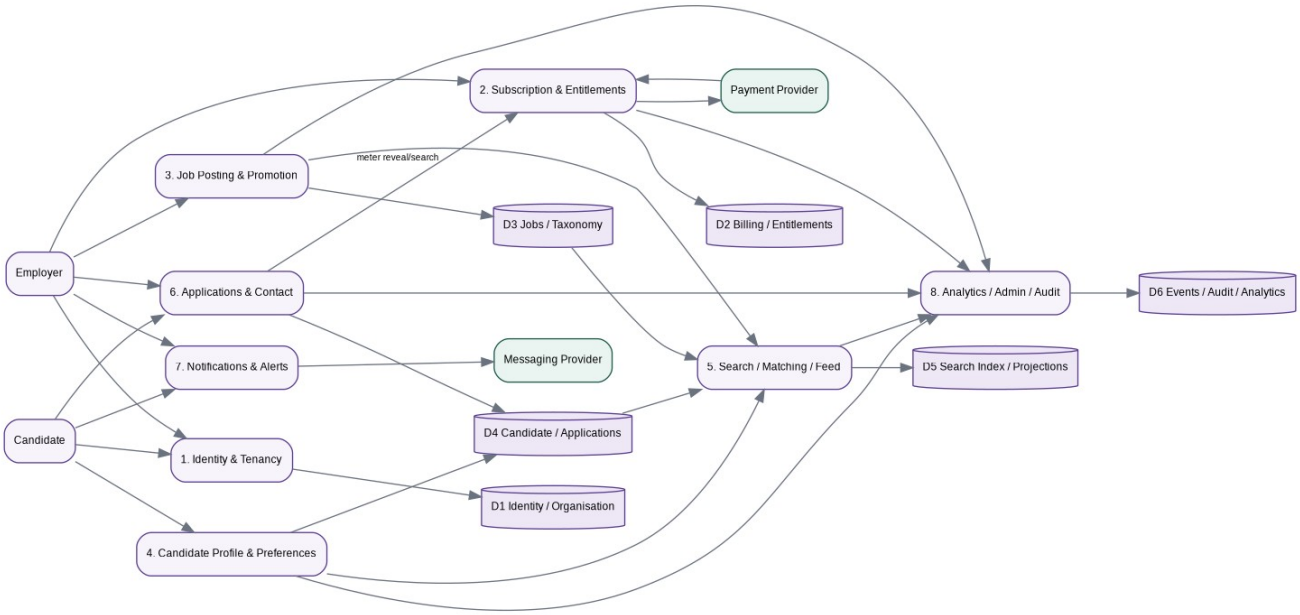
1. Architecture Principles

- SQL Server is the canonical transactional store; search projections/indexes are derived and rebuildable.
- GraphQL is product-facing and does not expose database tables directly.
- Organisation is the SaaS tenancy/billing boundary; UserAccount is the identity boundary.
- Subscription authorization is centralized in EntitlementService.
- Occupation taxonomy remains independent from collar market tags, industry, engagement and work mode.
- Candidate privacy is enforced before commercial entitlement; payment never overrides candidate consent.
- Eligibility filtering precedes semantic relevance and promotional boost.
- Write-side domain events use an outbox/inbox pattern for payment, search-index and messaging reliability.

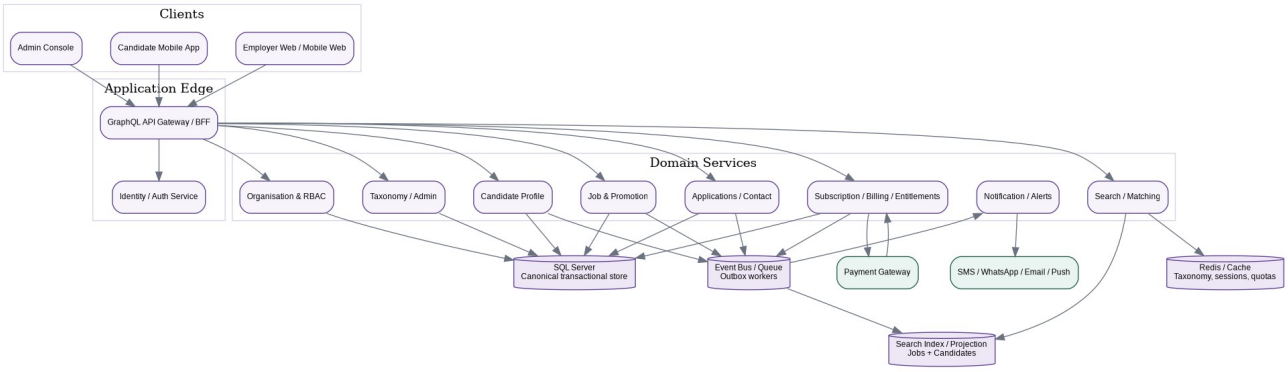
2. System Context / DFD Level 0



3. DFD Level 1



4. Logical Component Architecture



5. Domain Boundaries

Domain	Owns	Does not own
Identity	UserAccount, LoginIdentifier, OTP/session security.	Organisation subscription or candidate work profile.
Organisation	Organisation, memberships, identifiers, locations, RBAC.	Candidate profile.
Taxonomy	TaxonomyVersion, JobFamily, Role, Specialisation, WorkContext, Skill, CredentialType, AssetType.	Employer/candidate transactional state.
Billing	Plan, PlanVersion, Entitlements, Subscription, Usage, Invoice/Payment metadata, provider webhooks.	Job ranking or candidate consent.

Domain	Owns	Does not own
Jobs	JobPosting and structured requirements, compensation, schedule, contact policy.	Candidate employment history.
Promotion	PromotionProduct, JobPromotion, promotion attribution.	Eligibility truth.
Candidate	CandidateProfile, occupations, skills, credentials, assets, preferences, consent.	Organisation plan.
Applications	CandidateApplication, snapshots, answers, events, employer workflow.	Billing plan definitions.
Matching/Search	Derived JobSearchDocument/CandidateSearchDocument and ranking logic.	Canonical PII.
Notification	Alert subscriptions/preferences and delivery jobs.	Authentication truth.
Audit/Analytics	Access logs, event facts, operational metrics.	Business source-of-truth records.

6. Unified Database Relationship

```
UserAccount
  ├─ LoginIdentifier / OtpChallenge
  ├─ OrganisationMembership ── Organisation ── OrganisationSubscription
  │                          ├─ JobPosting ── JobPromotion
  │                          ├─ SubscriptionUsageEvent
  │                          └─ CandidateAccessAudit
  └─ CandidateProfile
     ├─ CandidateOccupation / Skill / Credential / Asset / Preference / Consent
     ├─ CandidateApplication ──────────────── JobPosting
     ├─ JobAlertSubscription
     └─ CandidateContactEvent

SubscriptionPlan -> SubscriptionPlanVersion -> PlanEntitlement
OrganisationSubscription -> UsageEvent / BillingInvoice / PaymentTransaction
JobPosting -> Role/Specialisation/WorkContext + requirement tables
CandidateProfile -> Candidate search projection (derived)
JobPosting -> Job search projection (derived)
```

7. SaaS/Billing Data Model

Entity	Key fields / purpose
SubscriptionPlan	Stable product identity: STARTER/PRO/BUSINESS etc; marketing name and availability.
SubscriptionPlanVersion	Immutable price/interval/config version: TRIAL/MONTHLY/YEARLY, currency, amount, effective dates.

Entity	Key fields / purpose
EntitlementDefinition	Feature/quota code such as ACTIVE_JOBS, CANDIDATE_SEARCH, CONTACT_REVEAL, RECRUITER_SEAT, PROMOTION_CREDIT, EXPORT, API_ACCESS.
PlanEntitlement	Limit/feature value for a plan version; reset cadence and enforcement mode.
OrganisationSubscription	Current subscription state, provider refs, trial/period dates, cancellation behavior.
EntitlementUsageEvent	Immutable metering event with quantity, reference, period and idempotency key.
PaymentProviderCustomer	Maps organisation to external billing customer id.
BillingInvoice	Provider invoice/order reference, total, currency, state, due/paid dates.
PaymentTransaction	Attempt/refund/payment evidence; no card secrets.
BillingWebhookInbox	Raw verified provider event metadata + processing state/idempotency.

8. Promotion Data Model

Entity	Purpose
PromotionProduct	Configurable placement type, duration, credit cost/price, max boost and eligibility policy.
JobPromotion	Job + product + start/end + status + source (included credit/add-on/admin).
PromotionAttributionEvent	Impression/view/apply/contact attribution for measurement.

9. Entitlement Evaluation

```
authorizeEmployerAction(orgId, userId, entitlementCode, quantity, reference):
  1. Verify active OrganisationMembership + RBAC permission.
  2. Resolve effective subscription (trial/paid/admin grant).
  3. Resolve immutable plan version + entitlement definition.
  4. Compute current usage window and remaining quantity.
  5. If feature disabled -> DENY.
  6. If hard numeric limit exceeded -> DENY + upgrade metadata.
  7. If action is meterable -> reserve/record usage using idempotency key.
  8. Write audit/usage event and return allowance + remaining/resetAt.
```

10. Candidate Access Order of Operations

```
Candidate visibility/consent
    ↓
Employer organisation membership/RBAC
    ↓
Subscription/entitlement
    ↓
```

```
Rate-limit / anti-abuse policy
    ↓
Mask or reveal fields
    ↓
CandidateAccessAudit + EntitlementUsageEvent
```

Privacy comes before monetisation: a paid plan never overrides candidate discoverability or consent.

11. Search and Matching Architecture

Projection	Contains	Excludes
JobSearchDocument	Job id, taxonomy ids, structured requirement facets, compensation, geo, freshness, promotion metadata.	Employer secrets, raw sensitive identifiers.
CandidateSearchDocument	Candidate id, discoverability flag, taxonomy roles, structured skills, coarse geo/radius, availability, verification flags.	Phone/email, raw Aadhaar/PAN, credential numbers, unnecessary PII.

Use CDC/outbox-driven projection updates. Search results return ids + ranking/match reasons; GraphQL hydrates authoritative details from SQL with batching/DataLoader. Cursor/keyset pagination is required for large result sets.

12. GraphQL Architecture

Surface	Representative operations
Public/identity	plans, requestCandidateOtp, verifyCandidateOtp, requestEmployerOtp, verifyEmployerOtp.
Employer SaaS	myOrganisations, subscription, entitlementUsage, startTrial, createCheckout, changePlan, cancelSubscription.
Employer jobs	createJobPosting, publishJob, pauseJob, closeJob, promoteJob, myJobs.
Employer talent	candidateSearch, jobCandidateMatches, candidateProfileMasked, revealCandidateContact, shortlistCandidate.
Candidate	meCandidate, jobFeed, jobSearch, prepareJobApplication, applyToJob, myApplications, alerts, contactEmployer.
Admin	plan/taxonomy/promotion configuration, moderation, reconciliation and audit queries through privileged schema/policies.

13. Event Catalog

Event	Producer	Consumers / purpose
USER_OTP_VERIFIED	Identity	Provision candidate/employer session; analytics.
ORGANISATION_CREATED	Organisation	Default settings; analytics.

Event	Producer	Consumers / purpose
SUBSCRIPTION_STARTED	Billing	Refresh entitlements; notifications.
SUBSCRIPTION_RENEWED	Billing	Open new usage window; notifications.
SUBSCRIPTION_PAST_DUE	Billing	Apply grace/restriction policy.
JOB_PUBLISHED	Jobs	Index job; recommendation fanout.
JOB_UPDATED	Jobs	Re-index job.
JOB_PROMOTION_STARTED	Promotion	Ranking metadata/index update.
CANDIDATE_PROFILE_UPDATED	Candidate	Re-index searchable candidate.
APPLICATION_SUBMITTED	Applications	Notify employer; metrics.
APPLICATION_STATUS_CHANGED	Applications	Notify candidate; metrics.
CANDIDATE_CONTACT_REVEALED	Talent access	Meter usage; audit.

14. Security and Privacy

- OTP challenges are hashed, short-lived, attempt-limited and rate-limited by destination/device/IP risk signals.
- Access tokens are short-lived; refresh tokens are revocable/rotatable. Employer requests require organisation membership context.
- Sensitive organisation/candidate identifier values are encrypted; searchable uniqueness uses keyed/hash representations where appropriate.
- No Aadhaar/PAN/credential numbers are copied into search projections or analytics events.
- Candidate contact reveal is purpose-limited, auditable and metered; bulk export is a distinct entitlement/policy.
- Payment card/bank secrets remain with the payment provider; store only provider tokens/ids and payment status metadata.
- Use row-level application authorization in services; do not rely on GraphQL field omission alone.
- Administrative overrides require explicit role, reason and audit record.

15. Non-Functional Requirements

NFR	Target / requirement
Availability	Target 99.9% for core API at MVP; isolate provider outages with queues/retries.
Latency	P95 feed/search <= 700 ms from warm region under target load; simple mutations <= 1 s excluding external payment/OTP providers.
Scalability	Stateless GraphQL/API tier; independent search workers; horizontal scale.
Consistency	Strong consistency for billing/entitlement meter, publish/apply/reveal; eventual consistency acceptable for search index/analytics.
Idempotency	Required for payment webhooks, apply,

NFR	Target / requirement
	reveal/meter, promotion purchase/activation and other retry-prone writes.
Observability	Correlation id; structured logs; metrics for auth, billing, GraphQL latency, search latency, queue lag and conversion funnels.
Recovery	Automated SQL backups/PITR per deployment environment; replayable outbox/inbox where feasible.
Data retention	Policy-driven by record type/purpose; audit/billing retention separate from discoverability.

16. Deployment Topology

- CDN/WAF -> GraphQL/API application tier.
- Identity/auth module/service and organisation/RBAC module.
- Billing/entitlement module with payment-provider adapter and webhook endpoint.
- Job/candidate/application domain modules; can begin as a modular monolith and split later by load/team boundaries.
- SQL Server primary transactional database.
- Redis for rate limits, taxonomy/entitlement cache and short-lived coordination.
- OpenSearch/Elasticsearch-compatible or equivalent search engine for job/candidate projections.
- Queue/event bus and background workers for indexing, notifications, billing reconciliation and analytics.
- Object storage for resume/evidence artifacts, never as the source of profile truth.

17. Recommended Delivery Phases

Phase	Scope
P0 - Foundation	Taxonomy v2 + unified identity + organisation + candidate profile; schema migrations; feature flags.
P1 - Marketplace MVP	Employer trial, job post/publish, candidate OTP/feed/apply, applications, basic billing status.
P2 - Talent SaaS	Candidate search projection, consent/discoverability, masked profiles, contact reveal + metering.
P3 - Monetisation	Monthly/yearly checkout, renewal webhooks, plan dashboard, add-ons, promotion credits.
P4 - Scale/Quality	Advanced matching explanations, moderation/fraud, analytics, ATS/API, richer verification.

18. Source Lineage

The database and GraphQL designs intentionally extend, rather than replace, IndiaWorks Employer/Auth/Job Posting v3 and Candidate Onboarding/Application v4. The associated .sql and .graphql deliverables in this v7 bundle contain the integrated starter contracts.

v7 commercial gate - five free candidates

Rule: A verified organisation/legal entity receives five lifetime free DISTINCT candidate unlocks. The sixth distinct candidate requires a paid monthly/yearly/add-on CANDIDATE_UNLOCK entitlement.

Rule	v7 behavior
Masked discovery	Applicant/search result cards and candidate counts are visible in masked form without consuming an unlock.
Consumption key	Only first full-profile access of a distinct (OrganisationId, CandidateProfileId) consumes candidate access.
Scope	The five free candidates are shared by every recruiter, every job and every candidate-access surface in the organisation.
No reset	Free allowance never resets by month, billing period, new recruiter, new job, device, cancellation or re-sign-up.
Duplicate candidate	Re-opening the same candidate or finding the same candidate against another job does not consume again.
Privacy first	Consent/application context, discoverability and candidate-organisation block rules are evaluated before payment/entitlement.
Paid access	Paid plan/add-on entitlements can grant further CANDIDATE_UNLOCK units or unlimited access.
Contact	Candidate contact reveal remains separately consent checked, entitlement checked and audited.
Abuse control	Commercial identity fingerprint/risk controls prevent recreating organisations purely to regain five free candidates.

20. v7 candidate-access database design

Entity	Key columns / purpose
OrganisationCommercialAccessState	OrganisationId PK, commercial identity fingerprint, FreeCandidateUnlockLimit=5, grant/revoke metadata, rowversion.
OrganisationCandidateUnlock	Unique OrganisationId+CandidateProfileId; source FREE_ALLOWANCE/PAID_ENTITLEMENT/ADD_ON/ADMIN; first job/user/time; usage-event link.
CandidatePaywallEvent	PAYWALL_SHOWN/CHECKOUT_STARTED/ PAYMENT_COMPLETED/UNLOCK_RETRIED with freeUsed/freeLimit and correlation id.
EntitlementDefinition	Add CANDIDATE_UNLOCK. FREE_EVALUATION plan uses limit 5, reset NONE.
EntitlementUsageEvent	Immutable usage record; unique org+entitlement+idempotency key; paid unlock usage and other metered actions.

Concurrency rule

Candidate unlock must execute inside an application-service transaction. First check unique OrganisationCandidateUnlock; then lock organisation commercial access/entitlement state, evaluate free/paid allowance, insert unlock and usage event atomically. The unique (OrganisationId, CandidateProfileId) constraint is the final database guard against duplicate consumption.

21. v7 candidate access order of operations

1. Authenticate employer session and resolve active organisation membership/RBAC.

2. Evaluate candidate privacy/application/discoverability and CandidateOrganisationBlock.

3. Check whether OrganisationCandidateUnlock already exists; if yes return full profile without metering.

4. Evaluate organisation free distinct-candidate usage. If below five, create FREE_ALLOWANCE unlock.

5. If five are already used, resolve active paid/add-on CANDIDATE_UNLOCK entitlement.

6. If paid entitlement allows, create PAID_ENTITLEMENT/ADD_ON unlock and usage event.

7. Otherwise write CandidatePaywallEvent(PAYWALL_SHOWN) and return PAYWALL_REQUIRED with plan choices.

8. Only after candidate is unlocked may contact reveal be evaluated; contact remains separately consent/entitlement audited.

22. GraphQL v7 extensions

Surface	Purpose
candidateAccessStatus(organisationId)	Free limit/used/remaining, paidAccessActive, paywallActive and candidate-unlock entitlement usage.
jobApplicantsV7	Masked applicant cards + access state + shared organisation access status.
unlockCandidateProfile	Single authoritative value-event mutation; returns FREE_UNLOCKED/PAID_UNLOCKED/PAYWALL_REQUIRED/etc.
MaskedCandidateCard.accessDecision	Allows UI to show already-unlocked/free/paywall state without exposing full profile.
revealCandidateContact	Separate post-unlock operation; privacy and CONTACT_REVEAL policy remain authoritative.

23. Search/filter architecture

Job filters and candidate filters are projection-backed. jobSearchFacets must not issue high-cardinality OLTP joins on each mobile filter change. Cache by normalized filter hash + geography + taxonomy version + index version. Candidate search projection excludes phone/email, raw Aadhaar/PAN and credential numbers.

Job filter family	Indexed fields

Taxonomy	familyId, roleId, specialisationId, workContextIds, aliases/search terms
Geo	city/district/PIN/lat/lng, distance/radius
Work	workMode, engagementType
Experience/Education	experience ranges/policy, education level, course
Pay	monthlyEquivalentMin/max, payFrequency, bonus/incentive flags
Schedule	shiftType, start/end, night/rotational
Freshness/Urgency	publishedAt, lastRefreshedAt, hiringUrgency
Capabilities	skills, languages, credential/asset requirements
Employer	verified flag, sourceType, sector, industry, employeeBand
Benefits/Contact	benefit codes, immediate call/WhatsApp capability

24. Session/persona architecture

A single UserAccount can own a CandidateProfile and multiple OrganisationMemberships. AuthSession activePersona is a session context, not a permanent identity type. UserSession persists refresh-token hash/device/persona and supports token rotation, logout, revoke-other-sessions and sensitive-identifier-change revocation.

25. Alerts, sharing and deep links

SavedJobSearch stores a versioned canonical JobSearchInput. JobAlertSubscription stores channel/frequency/quiet hours; JobAlertRun stores watermarks; JobAlertDelivery stores per-job/channel dedupe and provider status. JobPublicLink/JobShareLink uses stable HTTPS public code/slug and native app links/universal links with mobile-web fallback.

26. Employer-candidate CRM model

CandidateApplication is Candidate<->Job. EmployerCandidateRelationship is Organisation<->Candidate across jobs. OrganisationCandidateUnlock is commercial access. CandidateContactReveal is PII/contact access. CandidateOrganisationBlock is the safety boundary. Keeping these separate prevents subscription logic from corrupting hiring workflow semantics.

27. Release/dependency architecture

S0 Foundation -> S1 Identity/Tenancy/Free allowance provisioning -> S2 Billing/Entitlement/Five-free gate -> S3 Job Authoring -> S4 Candidate Discovery/Apply -> S5 Applicant CRM + Paywall -> S6 Paid Talent Search -> S7 Alerts/Promotion -> S8 Trust/Admin/Analytics -> S9 Hardening/UAT/Launch. The v7 backlog CSV has been topologically validated and priority-ordered inside each sprint.

28. v7 source lineage

Source	Integrated into v7
Unified SaaS v5	Two-sided SaaS, entitlements, promotion, unified employer/candidate architecture.

Product/Technical Review v6	Sessions/personas, filters/facets, alerts, deep links, CRM, trust/safety, moderation and SQL corrections.
v7 commercial decision	Five lifetime free distinct candidates per organisation/legal entity, sixth-candidate paywall, candidate-unlock entity/API and sprint dependency restructure.